

P.S.R. ENGINEERING COLLEGE, SIVAKASI – 626140.

(An Autonomous Institution – Affiliated to Anna University)

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

(ACADEMIC YEAR 2025-2026 ODD SEMESTER)

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CASE STUDY

Course Name : Data Structures **Year / Branch : II / CSE**

Course Code : 231CS32 **Semester III**

S.No.	Questions	Marks	Bloom's Category Level	Course Outcome
1.	Shopping List Store items in a shopping list. Add an item. Remove an item. Search for an item. Display all items. Write a C++ program to implement the case study.		Us	CO1
2.	A train consists of coaches connected together. Coach1 → Coach2 → Coach3 → Coach4 → NULL Write a program to: Add a coach at the beginning Add a coach at the end Remove a coach Display all coaches Write a C++ program to implement the case study.		Ap	CO2
3.	Cars enter a narrow parking garage. Cars can only leave in the reverse order in which they entered. Write a C++ program to implement this scenario.		Ap	CO2
4.	A browser records: google.com youtube.com abcsc.com facebook.com When the user presses Back, the most recently visited page is removed. If Back is pressed twice, what page is currently at the top of the history?. Write a C++ code.			**CO2**
5.	A warehouse records the weights of packages: Input:			**CO3**

	7 25 10 45 30 15 50 20 Use Merge Sort to arrange them in ascending order.			
6.	A bank uses a hash table to store account numbers. For an account number x: index = x % table_size If that position is occupied, use linear probing: (index + 1) % table_size Continue until an empty position is found. Implement: INSERT x SEARCH x DISPLAY			**CO4**
7.	An organization stores employee IDs in a BST. Given: 50 30 70 20 40 60 80			**CO5**
8.	A government wants to connect several cities with roads. Each possible road has a construction cost. Find the minimum cost to connect every city so that every city can reach every other city. Use Prim's Algorithm. Cities: 4 Roads: 1 2 10 1 3 6 1 4 5 2 4 15 3 4 4			**CO6**

Note: Rm-Remebering, Us-Understanding, Ap-Applying, Ay-Analyzing, Ev-Evaluating, Cr-Creating

Course Outcomes: The Students will be able to

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CO1:	Outline the basic data structures concepts.
CO2:	Rephrase basic data structures such as linked lists, stacks and queues.
CO3:	Apply algorithms for solving problems in sorting and searching.
CO4:	Summarize the pros and cons of hashing techniques.
CO5:	Explain the implementations and applications of non-linear data structures.
CO6:	Illustrate the representation of graphs and its various algorithms.

Prepared By
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Approved By
HOD/CSE